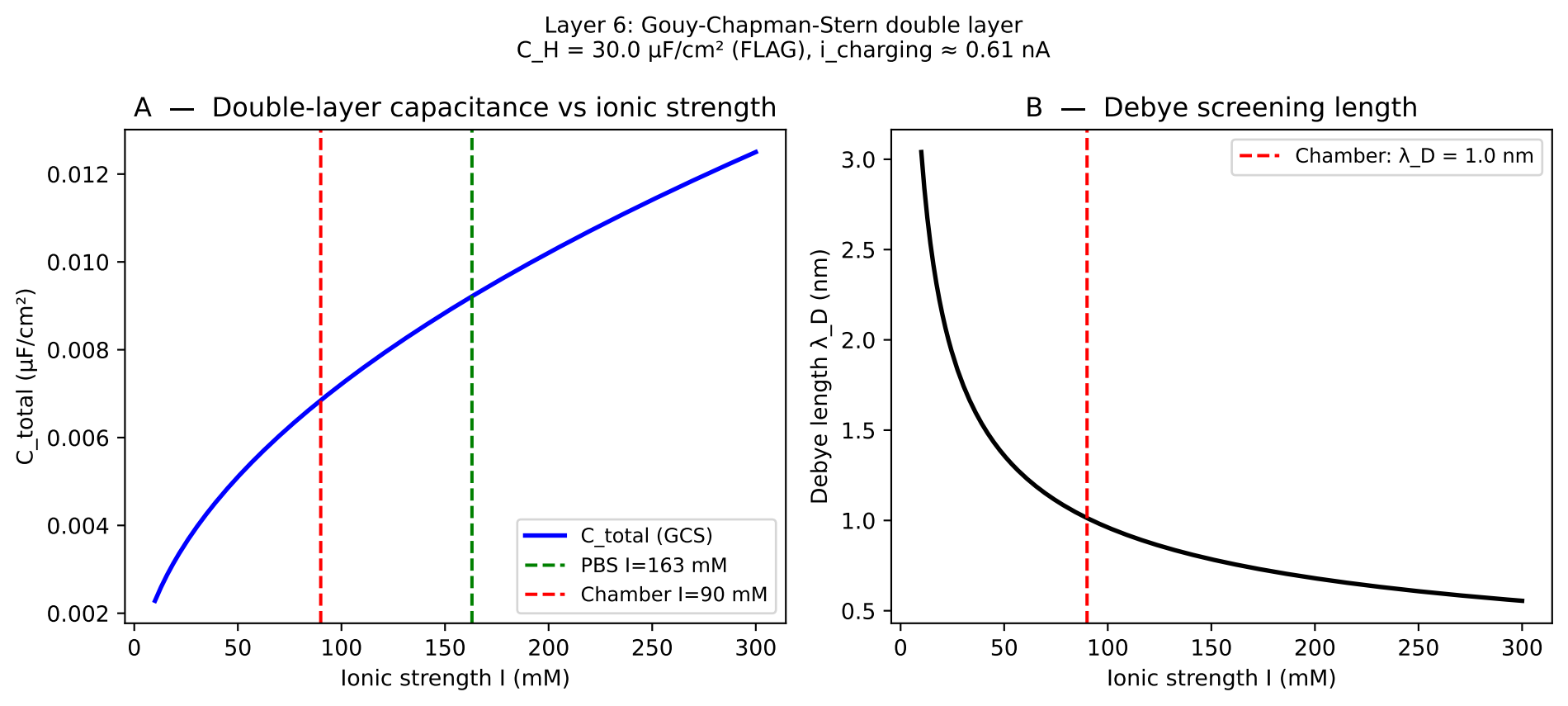
Layer 6: Gouy-Chapman-Stern double layer
C_H = 30.0 μF/cm² (FLAG), i_charging ≈ 0.61 nA
A — Double-layer capacitance vs ionic strength
C_total (μF/cm²)
0.002
0.004
0.006
0.008
0.010
0.012
0
50
100
150
200
250
300
Ionic strength I (mM)
C_total (GCS)
PBS I=163 mM
Chamber I=90 mM
B — Debye screening length
Debye length λ_D (nm)
0.5
1.0
1.5
2.0
2.5
3.0
Ionic strength I (mM)
Chamber: λ_D = 1.0 nm